

关于 $f(R)$ 引力

季皓洋

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1 基本定义与约定

1.1 $f(R)$ 理论

四维时空中 $f(R)$ 引力的度规表述下作用量为

$$S = \frac{1}{2\kappa} \int d^4x \sqrt{g} [f(R) - 2\Lambda] + S_{matter} \quad (1.1)$$

对度规做变分得到的场方程为

$$f'(R)R_{\mu\nu} - \left(\frac{1}{2}f(R) - \Lambda\right)g_{\mu\nu} - [\nabla_\mu \nabla_\nu - g_{\mu\nu} \square] f'(R) = \kappa T_{\mu\nu} \quad (1.2)$$

两边取迹得

$$f'(R)R - 2f(R) + 4\Lambda + 3\square f'(R) = \kappa T \quad (1.3)$$

若

$$f(R) = R + \alpha R^2 \quad (1.4)$$

则

$$-R + 4\Lambda + 6\alpha\square R = \kappa T \quad (1.5)$$

这看上去是一个关于 R 的 Klein-Gordon 方程. 但实际上由于 T 需要联立引力场方程和物质场方程才能解得, 即物质场与引力场存在耦合, 所以 T 实际上与 R 有关.

1.2 基本概念与定义

克里斯多弗符号为

$$\Gamma_{\mu\nu}^\rho = \frac{1}{2}g^{\rho\lambda}(\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu}) \quad (1.6)$$

黎曼曲率张量为

$$R^\sigma{}_{\rho\mu\nu} = \partial_\mu \Gamma_{\nu\rho}^\sigma - \partial_\nu \Gamma_{\mu\rho}^\sigma + \Gamma_{\mu\lambda}^\sigma \Gamma_{\nu\rho}^\lambda - \Gamma_{\nu\lambda}^\sigma \Gamma_{\mu\rho}^\lambda \quad (1.7)$$

里奇张量为

$$R_{\mu\nu} = R^\rho{}_{\mu\rho\nu} \quad (1.8)$$

达朗贝尔算符为

$$\square f = \nabla^2 f = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} g^{\mu\nu} \partial_\nu f) \quad (1.9)$$

1.3 外尔变换

外尔变换为

$$d\tilde{s}^2 = \tilde{g}_{\mu\nu} dx^\mu dx^\nu = e^{\Phi(x)} g_{\mu\nu} dx^\mu dx^\nu \quad (1.10)$$

做外尔变换后, 发生如下变化

$$\begin{aligned} \Gamma_{bc}^a(\tilde{g}) &= \Gamma_{bc}^a(g) + \frac{1}{2} (\delta_b^a \nabla_c \Phi + \delta_c^a \nabla_b \Phi - g_{bc} \nabla^a \Phi) \\ &\equiv \Gamma_{bc}^a(g) + \gamma_{bc}{}^a \end{aligned} \quad (1.11)$$

$$R_{\mu\nu}(\tilde{g}) = R_{\mu\nu}(g) - \frac{1}{2} [(D-2) \nabla_\mu \nabla_\nu \Phi + g_{\mu\nu} \square \Phi] - \frac{D-2}{4} [g_{\mu\nu} \nabla_\rho \Phi \nabla^\rho \Phi - \nabla_\mu \Phi \nabla_\nu \Phi] \quad (1.12)$$

$$R(\tilde{g}) = e^{-\Phi} \left[R(g) - (D-1) \square \Phi - \frac{(D-1)(D-2)}{4} \nabla_\rho \Phi \nabla^\rho \Phi \right] \quad (1.13)$$

其中 D 为时空维数, 设 $D = 4$, 则

$$R_{\mu\nu}(\tilde{g}) = R_{\mu\nu}(g) - \frac{1}{2} [2 \nabla_\mu \nabla_\nu \Phi + g_{\mu\nu} \square \Phi] - \frac{1}{2} [g_{\mu\nu} \nabla_\rho \Phi \nabla^\rho \Phi - \nabla_\mu \Phi \nabla_\nu \Phi] \quad (1.14)$$

$$R(\tilde{g}) = e^{-\Phi} \left[R(g) - 3 \square \Phi - \frac{3}{2} \nabla_\rho \Phi \nabla^\rho \Phi \right] \quad (1.15)$$

这里 $R(\tilde{g}) = \tilde{g}^{\mu\nu} R_{\mu\nu}(\tilde{g})$.

外尔变换的另一种常见的形式可以通过做代换 $\Omega^2(x) = e^{\Phi(x)}$, 即

$$\Phi = 2 \ln \Omega \quad (1.16)$$

则

$$\nabla_\mu \Phi = 2 \frac{1}{\Omega} \nabla_\mu \Omega \quad (1.17)$$

$$\square\Phi = 2(-\frac{1}{\Omega^2})\nabla^\mu\Omega\nabla_\mu\Omega + 2\frac{1}{\Omega}\square\Omega \quad (1.18)$$

$$\nabla_\mu\nabla_\nu\Phi = 2(-\frac{1}{\Omega^2})\nabla_\nu\Omega\nabla_\mu\Omega + 2\frac{1}{\Omega}\nabla_\mu\nabla_\nu\Omega \quad (1.19)$$

$$\nabla\Phi \cdot \nabla\Phi = 4\frac{1}{\Omega^2}\nabla_\mu\Omega\nabla^\mu\Omega \quad (1.20)$$

$$2\square\Phi + \nabla\Phi \cdot \nabla\Phi = 4\frac{1}{\Omega}\square\Omega \quad (1.21)$$

$$\begin{aligned} R(\tilde{g}) &= \Omega^{-2} \left[R(g) - 3 \left(2(-\frac{1}{\Omega^2})\nabla^\mu\Omega\nabla_\mu\Omega + 2\frac{1}{\Omega}\square\Omega \right) - 6\frac{1}{\Omega^2}\nabla\Omega \cdot \nabla\Omega \right] \\ &= \Omega^{-2} \left[R(g) - 6\frac{1}{\Omega}\square\Omega \right] \end{aligned} \quad (1.22)$$

$$\begin{aligned} R_{\mu\nu}(\tilde{g}) &= R_{\mu\nu}(g) \\ &- \frac{1}{2} \left[-4\Omega^{-2}\nabla_\mu\Omega\nabla_\nu\Omega + 4\Omega^{-1}\nabla_\mu\nabla_\nu\Omega + g_{\mu\nu}(-2\Omega^{-2}\nabla^\mu\Omega\nabla_\mu\Omega + 2\Omega^{-1}\square\Omega) \right] \\ &- \frac{1}{2} \left[4\Omega^{-2}g_{\mu\nu}\nabla\Omega \cdot \nabla\Omega - 4\Omega^{-2}\nabla_\mu\Omega\nabla_\nu\Omega \right] \\ &= R_{\mu\nu}(g) \\ &+ 2\Omega^{-2}\nabla_\mu\Omega\nabla_\nu\Omega - 2\Omega^{-1}\nabla_\mu\nabla_\nu\Omega + g_{\mu\nu}(\Omega^{-2}\nabla^\mu\Omega\nabla_\mu\Omega - \Omega^{-1}\square\Omega) \\ &- 2\Omega^{-2}g_{\mu\nu}\nabla\Omega \cdot \nabla\Omega + 2\Omega^{-2}\nabla_\mu\Omega\nabla_\nu\Omega \\ &= R_{\mu\nu}(g) + 4\Omega^{-2}\nabla_\mu\Omega\nabla_\nu\Omega - 2\Omega^{-1}\nabla_\mu\nabla_\nu\Omega + g_{\mu\nu}(-\Omega^{-2}\nabla^\mu\Omega\nabla_\mu\Omega - \Omega^{-1}\square\Omega) \end{aligned} \quad (1.23)$$

$$\gamma_{bc}{}^a = \Omega^{-1}(\delta_b^a\nabla_c\Omega + \delta_c^a\nabla_b\Omega - g_{bc}\nabla^a\Omega) \quad (1.24)$$

若满足文献中提到的受限外尔变换 (restricted Weyl transformation) 条件

$$\square\Omega = 0 \quad (1.25)$$

则有

$$R(\tilde{g}) = \Omega^{-2}R(g) \quad (1.26)$$

2 弱场极限

弱场极限下度规取

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \quad (2.1)$$

并记

$$h = h_{\mu\nu}\eta^{\mu\nu} \quad (2.2)$$

于是弱场极限下

$$\Gamma_{\mu\nu}^\rho = \frac{1}{2}\eta^{\rho\lambda}(\partial_\mu h_{\nu\lambda} + \partial_\nu h_{\mu\lambda} - \partial_\lambda h_{\mu\nu}) \quad (2.3)$$

$$R_{\mu\nu} \sim \partial_\mu \Gamma_{\nu\rho}^\sigma - \partial_\nu \Gamma_{\mu\rho}^\sigma \sim \frac{1}{2}(\partial_\mu \partial^\rho h_{\rho\nu} + \partial_\nu \partial^\rho h_{\rho\mu} - \partial_\mu \partial_\nu h - \partial_\rho \partial^\rho h_{\mu\nu}) \quad (2.4)$$

$$R \sim \eta^{\mu\nu} R_{\mu\nu} = \partial^\mu \partial^\nu h_{\mu\nu} - \partial^\mu \partial_\mu h \quad (2.5)$$

我们假设

$$f(R) = R + \alpha R^2 + \mathcal{O}(R^3) \quad (2.6)$$

则

$$\nabla_\nu f'(R) \sim 2\alpha \partial_\mu R \quad (2.7)$$

$$\nabla_\mu \nabla_\nu f'(R) \sim 2\alpha(\partial_\mu \partial_\nu R - \Gamma_{\mu\nu}^\rho \partial_\rho R) \sim 2\alpha \partial_\mu \partial_\nu (\partial^\rho \partial^\sigma h_{\rho\sigma} - \partial^\rho \partial_\rho h) \quad (2.8)$$

$$\square f'(R) \sim 2\alpha \partial^a \partial_a (\partial^\rho \partial^\sigma h_{\rho\sigma} - \partial^\rho \partial_\rho h) \quad (2.9)$$

将以上代入(1.2)并只保留线性项, 得到

$$\begin{aligned} & \frac{1}{2}(\partial_\mu \partial^\rho h_{\rho\nu} + \partial_\nu \partial^\rho h_{\rho\mu} - \partial_\mu \partial_\nu h - \partial_\rho \partial^\rho h_{\mu\nu}) - \frac{1}{2}\eta_{\mu\nu}(\partial^\rho \partial^\sigma h_{\rho\sigma} - \partial^\rho \partial_\rho h) \\ & - 2\alpha[\partial_\mu \partial_\nu - \eta_{\mu\nu} \partial^a \partial_a](\partial^\rho \partial^\sigma h_{\rho\sigma} - \partial^\rho \partial_\rho h) + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu} \end{aligned} \quad (2.10)$$

记

$$D_{\mu\nu} = \partial_\mu \partial_\nu - \eta_{\mu\nu} \partial^a \partial_a \quad (2.11)$$

则场方程整理为

$$\begin{aligned} & \partial_\mu \partial^\rho h_{\rho\nu} + \partial_\nu \partial^\rho h_{\rho\mu} - \partial_\rho \partial^\rho h_{\mu\nu} - \eta_{\mu\nu} \partial^\rho \partial^\sigma h_{\rho\sigma} - D_{\mu\nu} h - 4\alpha D_{\mu\nu} (\partial^\rho \partial^\sigma h_{\rho\sigma} - \partial^\rho \partial_\rho h) \\ & + 2\Lambda g_{\mu\nu} = 2\kappa T_{\mu\nu} \end{aligned} \quad (2.12)$$

这里存在规范冗余

$$h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu \quad (2.13)$$

取规范

$$\partial^\rho h_{\mu\rho} = 0 \quad (2.14)$$

则得到方程

$$\partial_\rho \partial^\rho h_{\mu\nu} - 2\Lambda g_{\mu\nu} + D_{\mu\nu} h - 4\alpha D_{\mu\nu} \partial^\rho \partial_\rho h = -2\kappa T_{\mu\nu} \quad (2.15)$$

两边取迹得

$$\square h - 6\alpha \square^2 h + \Lambda h + 4\Lambda = \kappa T \quad (2.16)$$

若 $\Lambda = 0$ 可做换元

$$\Phi = \square h \quad (2.17)$$

则有方程

$$\square \Phi - \frac{1}{6\alpha} \Phi = -\frac{1}{6\alpha} \kappa T \quad (2.18)$$

这是一个有源的 Klein-Gordon 方程.

3 微扰方法

设

$$g_{\mu\nu} = \sum_{n=0} \lambda^n g_{\mu\nu}^{(n)} \quad (3.1)$$

以及

$$T_{\mu\nu} = \sum_{n=0} \lambda^n T_{\mu\nu}^{(n)} \quad (3.2)$$

则

$$\Gamma_{\mu\nu}^\rho = \frac{1}{2} \sum_{n=0} \sum_{m=0} \lambda^{m+n} g^{(n)\rho\lambda} (g_{\lambda\nu,\mu}^{(m)} + g_{\lambda\mu,\nu}^{(m)} - g_{\mu\nu,\lambda}^{(m)}) \quad (3.3)$$

也可写为

$$\Gamma_{\mu\nu}^\rho = \sum_{n=0} \lambda^n \Gamma_{\mu\nu}^{(n)\rho} \quad (3.4)$$

其中

$$\Gamma^{(m+n)\rho}_{\mu\nu} = \frac{1}{2} \sum_{k=0}^{m+n} g^{(k)\rho\lambda} (g^{(m+n-k)}_{\lambda\nu,\mu} + g^{(m+n-k)}_{\lambda\mu,\nu} - g^{(m+n-k)}_{\mu\nu,\lambda}) \quad (3.5)$$

于是

$$\begin{aligned} \nabla_\mu \nabla_\nu f &= \partial_\mu \partial_\nu f - \Gamma^\rho_{\mu\nu} \partial_\rho f \\ &= \partial_\mu \partial_\nu f - \sum_{n=0} \lambda^n \Gamma^{(n)\rho}_{\mu\nu} \partial_\rho f \end{aligned} \quad (3.6)$$

$$\square f = \sum_{m=0} \lambda^m g^{(m)\mu\nu} \partial_\mu \partial_\nu f - \sum_{m=0} \sum_{n=0} \lambda^{n+m} g^{(m)\mu\nu} \Gamma^{(n)\rho}_{\mu\nu} \partial_\rho f \quad (3.7)$$

$$R^\sigma_{\rho\mu\nu} = \sum_{n=0} \lambda^n \left(\Gamma^{(n)\sigma}_{\nu\rho,\mu} - \Gamma^{(n)\sigma}_{\mu\rho,\nu} + \sum_{k=0}^n \left(\Gamma^{(k)\sigma}_{\mu\lambda} \Gamma^{(n-k)\lambda}_{\nu\rho} - \Gamma^{(k)\sigma}_{\nu\lambda} \Gamma^{(n-k)\lambda}_{\mu\rho} \right) \right) \quad (3.8)$$

也可记为

$$R^\sigma_{\rho\mu\nu} = \sum_{n=0} \lambda^n R^{(n)\sigma}_{\rho\mu\nu} \quad (3.9)$$

其余量类似. 设

$$f(R) = R + \lambda F(R) \quad (3.10)$$

于是

$$F(R) = \sum_{n=0} \lambda^n F^{(n)} \quad (3.11)$$

$$F'(R) = \sum_{n=0} \lambda^n F'^{(n)} \quad (3.12)$$

$$f(R) = \sum_{n=0} \lambda^n \left(R^{(n)} + F^{(n-1)} \right) \quad (3.13)$$

$$f'(R) = 1 + \sum_{n=0} \lambda^{n+1} \left(F'^{(n)} \right) \quad (3.14)$$

以上代入(1.2)得到

$$\begin{aligned}
& \left(1 + \sum_{n=0} \lambda^{n+1} \left(F'^{(n)}\right)\right) \sum_{m=0} \lambda^m R^{(m)}_{\mu\nu} - \frac{1}{2} \sum_{n=0} \lambda^n \left(R^{(n)} + F^{(n-1)}\right) \left(\sum_{m=0} \lambda^m g^{(m)}_{\mu\nu}\right) \\
& + \Lambda \sum_{n=0} \lambda^n g^{(n)}_{\mu\nu} \\
& - \left[\partial_\mu \partial_\nu - \sum_{m=0} \lambda^m \Gamma^{(m)\rho}_{\mu\nu} \partial_\rho - \sum_{m=0} \lambda^m g^{(m)}_{\mu\nu} \left(\sum_{r=0} \lambda^r g^{(r)\mu\nu} \partial_\mu \partial_\nu - \sum_{r=0} \sum_{s=0} \lambda^{r+s} g^{(r)\mu\nu} \Gamma^{(s)\rho}_{\mu\nu} \partial_\rho \right) \right] \\
& \left(\sum_{n=0} \lambda^{n+1} \left(F'^{(n)}\right) \right) = \kappa \sum_{n=0} \lambda^n T^{(n)}_{\mu\nu}
\end{aligned} \tag{3.15}$$

第 k 阶方程为

$$\begin{aligned}
& R^{(k)}_{\mu\nu} + \sum_{i=0}^k F'^{(i)} R^{(k-i-1)}_{\mu\nu} - \frac{1}{2} \sum_{i=0}^k (R^{(i)} + F^{(i-1)}) g^{(k-i)}_{\mu\nu} + \Lambda g^{(k)}_{\mu\nu} - \partial_\mu \partial_\nu F'^{(k-1)} \\
& + \sum_{i=0}^k \Gamma^{(i)\rho}_{\mu\nu} \partial_\rho F'^{(k-i-1)} + \sum_{\substack{i+j \leq k \\ i,j \geq 0}} g^{(i)}_{\mu\nu} g^{(j)ab} \partial_a \partial_b F'^{(k-i-j-1)} \\
& - \sum_{\substack{i+j+h \leq k \\ i,j,h \geq 0}} g^{(i)}_{\mu\nu} g^{(j)ab} \Gamma^{(h)\rho}_{ab} \partial_\rho F'^{(k-i-j-h-1)} = \kappa T^{(k)}_{\mu\nu}
\end{aligned} \tag{3.16}$$

特别地, 零阶方程

$$R^{(0)}_{\mu\nu} - \frac{1}{2} R^{(0)} g^{(0)}_{\mu\nu} + \Lambda g^{(0)}_{\mu\nu} = \kappa T^{(0)}_{\mu\nu} \tag{3.17}$$

是爱因斯坦场方程. 而一阶方程为

$$\begin{aligned}
& R^{(1)}_{\mu\nu} + F'^{(0)} R^{(0)}_{\mu\nu} - \frac{1}{2} R^{(0)} g^{(1)}_{\mu\nu} - \frac{1}{2} \left(R^{(1)} + F^{(0)} \right) g^{(0)}_{\mu\nu} + \Lambda g^{(1)}_{\mu\nu} - \partial_\mu \partial_\nu F'^{(0)} \\
& + \Gamma^{(0)\rho}_{\mu\nu} \partial_\rho F'^{(0)} + g^{(0)}_{\mu\nu} g^{(0)ab} \partial_a \partial_b F'^{(0)} - g^{(0)}_{\mu\nu} g^{(0)ab} \Gamma^{(0)\rho}_{ab} \partial_\rho F'^{(0)} = \kappa T^{(1)}_{\mu\nu}
\end{aligned} \tag{3.18}$$

类似地, 对方程(1.3)可进行类似地处理, 得到

$$\begin{aligned}
& (1 + \sum_{n=0} \lambda^n F'^{(n-1)}) \sum_{m=0} \lambda^m R^{(m)} - 2 \left(\sum_{n=0} \lambda^n (R^{(n)} + F^{(n-1)}) \right) + 4\Lambda \\
& + 3 \left(\sum_{m=0} \lambda^m g^{(m)\mu\nu} \partial_\mu \partial_\nu - \sum_{m=0} \sum_{n=0} \lambda^{n+m} g^{(m)\mu\nu} \Gamma_{\mu\nu}^{(n)\rho} \partial_\rho \right) \left(\sum_{n=0} \lambda^n F'^{(n-1)} \right) = \kappa \sum_{n=0} \lambda^n T^{(n)}
\end{aligned} \tag{3.19}$$

第 k 阶方程为

$$\begin{aligned}
& R^{(k)} + \sum_{i=0}^k R^{(i)} F'^{(k-i-1)} - 2R^{(k)} - 2F^{(k-1)} + 4\delta_{0k}\Lambda + 3 \sum_{i=0}^k g^{(i)ab} \partial_a \partial_b F'^{(k-i-1)} \\
& - 3 \sum_{\substack{i,j \geq 0 \\ i+j \leq k}} g^{(i)ab} \Gamma_{ab}^{(j)\rho} \partial_\rho F'^{(k-i-j-1)} = \kappa T^{(k)}
\end{aligned} \tag{3.20}$$

故零阶方程为

$$-R^{(0)} + 4\Lambda = \kappa T^{(0)} \tag{3.21}$$

这与 GR 一致. 而一阶方程为

$$-R^{(1)} + R^{(0)} F'^{(0)} - 2F^{(0)} + 3g^{(0)ab} \partial_a \partial_b F'^{(0)} - 3g^{(0)ab} \Gamma_{ab}^{(0)\rho} \partial_\rho F'^{(0)} = \kappa T^{(1)} \tag{3.22}$$

现在, 考虑

$$f(R) = R + \lambda R^2 \tag{3.23}$$

则

$$F'^{(n)} = 2R^{(n)} \tag{3.24}$$

$$F^{(n)} = \sum_{i=0}^n R^{(i)} R^{(n-i)} \tag{3.25}$$

此时一阶微扰方程为

$$\begin{aligned}
& R_{\mu\nu}^{(1)} + 2R^{(0)} R_{\mu\nu}^{(0)} - \frac{1}{2} R^{(0)} g_{\mu\nu}^{(1)} - \frac{1}{2} \left(R^{(1)} + R^{(0)} R^{(0)} \right) g_{\mu\nu}^{(0)} + \Lambda g_{\mu\nu}^{(1)} - 2\partial_\mu \partial_\nu R^{(0)} \\
& + 2\Gamma_{\mu\nu}^{(0)\rho} \partial_\rho R^{(0)} + 2g_{\mu\nu}^{(0)} g^{(0)ab} \partial_a \partial_b R^{(0)} - 2g_{\mu\nu}^{(0)} g^{(0)ab} \Gamma_{ab}^{(0)\rho} \partial_\rho R^{(0)} = \kappa T_{\mu\nu}^{(1)}
\end{aligned} \tag{3.26}$$

$$-R^{(1)} + 2R^{(0)}R^{(0)} - 2R^{(0)}R^{(0)} + 6g^{(0)ab}\partial_a\partial_b R^{(0)} - 6g^{(0)ab}\Gamma_{ab}^{(0)\rho}\partial_\rho R^{(0)} = \kappa T^{(1)} \quad (3.27)$$

若 $R^{(0)}$ 是常数, 则有

$$R_{\mu\nu}^{(1)} + 2R^{(0)}R_{\mu\nu}^{(0)} - \frac{1}{2}R^{(0)}g_{\mu\nu}^{(1)} - \frac{1}{2}\left(R^{(1)} + R^{(0)}R^{(0)}\right)g_{\mu\nu}^{(0)} + \Lambda g_{\mu\nu}^{(1)} = \kappa T_{\mu\nu}^{(1)} \quad (3.28)$$

$$-R^{(1)} = \kappa T^{(1)} \quad (3.29)$$

如果加上假设 $T^{(n)}$ 在任意阶都是常数, 那么不论计算多少阶微扰, 最终得到的里奇标量 R 永远是常数.

4 大 R 极限

4.1 $f(R) = R + \alpha R^2$

大 R 极限下

$$f(R) \sim \alpha R^2 \quad (4.1)$$

为了方便, 下面略去 α . 代入(1.2)和(1.3)得

$$2RR_{\mu\nu} - \left(\frac{1}{2}R^2 - \Lambda\right)g_{\mu\nu} - 2[\nabla_\mu\nabla_\nu - g_{\mu\nu}\square]R = \kappa T_{\mu\nu} \quad (4.2)$$

$$4\Lambda + 6\square R = \kappa T \quad (4.3)$$

当 $\kappa T = 4\Lambda$ 时, 得到 R 满足方程

$$\square R = 0 \quad (4.4)$$

这说明 R 是调和的, 并且(4.2) 化为

$$2RR_{\mu\nu} - \left(\frac{1}{2}R^2 - \Lambda\right)g_{\mu\nu} - 2\nabla_\mu\nabla_\nu R = \kappa T_{\mu\nu} \quad (4.5)$$

现在, 做受限外尔变换

$$\tilde{g}_{\mu\nu} = \Omega^2(x)g_{\mu\nu} \quad (4.6)$$

则根据(1.26)有

$$R(\tilde{g}) = \Omega^{-2}R(g) \quad (4.7)$$

于是作用量的引力部分

$$\int d^4x \sqrt{g} R^2 \quad (4.8)$$

在受限外尔变换下保持不变, 这一结果是四维时空的特例. 注意这里受限外尔变换未必保持运动方程不变, 比如

$$\begin{aligned} \square R(\tilde{g}) &= \nabla^\mu (-2\Omega^{-3}(\nabla_\mu \Omega)R(g) + \Omega^{-2}\nabla_\mu R(g)) \\ &= 6\Omega^{-4}(\nabla\Omega \cdot \nabla\Omega)R(g) - 4\Omega^{-3}(\nabla\Omega \cdot \nabla R(g)) \end{aligned} \quad (4.9)$$

不一定是零. 现在设 $\Omega = R$, 则有

$$R(\tilde{g}) = \frac{1}{R(g)} \quad (4.10)$$

则在大 R 极限下, $R(\tilde{g}) \rightarrow 0$. 因此可称 $f(R) = R + \alpha R^2$ 理论在大 R 极限下是共形标量平坦的 (我编的词, 不用在意). 该受限外尔变换实际上也是里奇标量的倒代换. 为了方便, 下面记 $\tilde{R} = R(\tilde{g})$, $\tilde{R}_{\mu\nu} = R_{\mu\nu}(\tilde{g})$. 于是

$$0 = \square R(g) = \square \frac{1}{\tilde{R}} = \nabla^\mu \left(-\tilde{R}^{-2} \nabla_\mu \tilde{R} \right) = 2\tilde{R}^{-3} \nabla^\mu R \nabla_\mu R - \tilde{R}^{-2} \square \tilde{R} \quad (4.11)$$

假设 $\tilde{R} \neq 0$,

$$\Rightarrow \tilde{R} \square \tilde{R} = 2 \nabla \tilde{R} \cdot \nabla \tilde{R} \quad (4.12)$$

另外, 由于

$$\tilde{R}_{\mu\nu} = R_{\mu\nu} + 4R^{-2} \nabla_\mu R \nabla_\nu R - 2R^{-1} \nabla_\mu \nabla_\nu R - g_{\mu\nu} R^{-2} \nabla R \cdot \nabla R \quad (4.13)$$

且

$$4R^{-2} \nabla_\mu R \nabla_\nu R = 4\tilde{R}^{-2} \nabla_\mu \tilde{R} \nabla_\nu \tilde{R} \quad (4.14)$$

$$\nabla_\mu \nabla_\nu R = \nabla_\mu \left(-\tilde{R}^{-2} \nabla_\nu \tilde{R} \right) = 2\tilde{R}^{-3} \nabla_\mu \tilde{R} \nabla_\nu \tilde{R} - \tilde{R}^{-2} \nabla_\mu \nabla_\nu \tilde{R} \quad (4.15)$$

$$-2R^{-1} \nabla_\mu \nabla_\nu R = -4\tilde{R}^{-2} \nabla_\mu \tilde{R} \nabla_\nu \tilde{R} + 2\tilde{R}^{-1} \nabla_\mu \nabla_\nu \tilde{R} \quad (4.16)$$

$$-g_{\mu\nu} R^{-2} \nabla R \cdot \nabla R = -g_{\mu\nu} \tilde{R}^{-2} \nabla \tilde{R} \cdot \nabla \tilde{R} \quad (4.17)$$

于是

$$\tilde{R}_{\mu\nu} = R_{\mu\nu} + 2\tilde{R}^{-1} \nabla_\mu \nabla_\nu \tilde{R} - g_{\mu\nu} \tilde{R}^{-2} \nabla \tilde{R} \cdot \nabla \tilde{R} \quad (4.18)$$

也即

$$R_{\mu\nu} = \tilde{R}_{\mu\nu} - 2\tilde{R}^{-1}\nabla_\mu\nabla_\nu\tilde{R} + g_{\mu\nu}\tilde{R}^{-2}\nabla\tilde{R}\cdot\nabla\tilde{R} \quad (4.19)$$

以上代入(4.5)得到

$$\begin{aligned} \kappa T_{\mu\nu} &= 2RR_{\mu\nu} - \left(\frac{1}{2}R^2 - \Lambda\right)g_{\mu\nu} - 2\nabla_\mu\nabla_\nu R \\ &= 2\tilde{R}^{-1}\left(\tilde{R}_{\mu\nu} - 2\tilde{R}^{-1}\nabla_\mu\nabla_\nu\tilde{R} + g_{\mu\nu}\tilde{R}^{-2}\nabla\tilde{R}\cdot\nabla\tilde{R}\right) - \left(\frac{1}{2}\tilde{R}^{-2} - \Lambda\right)g_{\mu\nu} \\ &\quad - 2\left(2\tilde{R}^{-3}\nabla_\mu\tilde{R}\nabla_\nu\tilde{R} - \tilde{R}^{-2}\nabla_\mu\nabla_\nu\tilde{R}\right) \end{aligned} \quad (4.20)$$

$\Rightarrow$

$$\begin{aligned} &2\tilde{R}^2\tilde{R}_{\mu\nu} - 4\tilde{R}\nabla_\mu\nabla_\nu\tilde{R} + 2g_{\mu\nu}\nabla\tilde{R}\cdot\nabla\tilde{R} - \left(\frac{1}{2}\tilde{R} - \Lambda\tilde{R}^3\right)g_{\mu\nu} \\ &- 4\nabla_\mu\tilde{R}\nabla_\nu\tilde{R} + 2\tilde{R}\nabla_\mu\nabla_\nu\tilde{R} = \kappa\tilde{R}^3T_{\mu\nu} \end{aligned} \quad (4.21)$$

$\Rightarrow$

$$2\tilde{R}^2\tilde{R}_{\mu\nu} - 2\tilde{R}\nabla_\mu\nabla_\nu\tilde{R} + 2g_{\mu\nu}\nabla\tilde{R}\cdot\nabla\tilde{R} - \left(\frac{1}{2}\tilde{R} - \Lambda\tilde{R}^3\right)g_{\mu\nu} - 4\nabla_\mu\tilde{R}\nabla_\nu\tilde{R} = \kappa\tilde{R}^3T_{\mu\nu} \quad (4.22)$$

为了让变换做得彻底些, 记 $\nabla_\mu(\tilde{g}) = \tilde{\nabla}_\mu$, $\tilde{g}^{\mu\nu}\tilde{\nabla}_\mu\tilde{\nabla}_\nu = \tilde{\square}$, $\tilde{g}^{\mu\nu}\tilde{\nabla}_\mu f\tilde{\nabla}_\nu f = \tilde{\nabla}f\cdot\tilde{\nabla}f$. 于是

$$\tilde{\nabla}_\mu f = \nabla_\mu f \quad (4.23)$$

$$\nabla_\mu\nabla_\nu f = \tilde{\nabla}_\mu\tilde{\nabla}_\nu f + \gamma_{\mu\nu}{}^a\tilde{\nabla}_a f \quad (4.24)$$

$$\nabla f\cdot\nabla f = \Omega^2\tilde{\nabla}f\cdot\tilde{\nabla}f \quad (4.25)$$

$$\begin{aligned} &2\tilde{R}^2\tilde{R}_{\mu\nu} - 2\tilde{R}\tilde{\nabla}_\mu\tilde{\nabla}_\nu\tilde{R} - 2\tilde{R}\gamma_{\mu\nu}{}^a\tilde{\nabla}_a\tilde{R} + 2\tilde{g}_{\mu\nu}\tilde{\nabla}\tilde{R}\cdot\tilde{\nabla}\tilde{R} \\ &- \left(\frac{1}{2}\tilde{R} - \Lambda\tilde{R}^3\right)\tilde{R}^2\tilde{g}_{\mu\nu} - 4\tilde{\nabla}_\mu\tilde{R}\tilde{\nabla}_\nu\tilde{R} = \kappa\tilde{R}^3T_{\mu\nu} \end{aligned} \quad (4.26)$$

$$\begin{aligned} &\tilde{R}^{-1}\left(\tilde{\square} + \tilde{g}^{\mu\nu}\gamma_{\mu\nu}{}^a\tilde{\nabla}_a\right)\tilde{R} = 2\tilde{R}^{-2}\tilde{\nabla}\tilde{R}\cdot\tilde{\nabla}\tilde{R} \\ &\Rightarrow \tilde{R}\left(\tilde{\square} + \tilde{g}^{\mu\nu}\gamma_{\mu\nu}{}^a\tilde{\nabla}_a\right)\tilde{R} = 2\tilde{\nabla}\tilde{R}\cdot\tilde{\nabla}\tilde{R} \end{aligned} \quad (4.27)$$

其中

$$\begin{aligned} \gamma_{\mu\nu}{}^a &= R^{-1}\left(\delta_\mu^a\nabla_\nu R + \delta_\nu^a\nabla_\mu R - g_{\mu\nu}g^{ab}\nabla_b R\right) \\ &= -\tilde{R}^{-1}\left(\delta_\mu^a\tilde{\nabla}_\nu\tilde{R} + \delta_\nu^a\tilde{\nabla}_\mu\tilde{R} - \tilde{g}_{\mu\nu}\tilde{g}^{ab}\tilde{\nabla}_b\tilde{R}\right) \end{aligned} \quad (4.28)$$

记 $\gamma^a = \tilde{g}^{\mu\nu} \gamma_{\mu\nu}^a$, 则

$$\gamma^a = \tilde{g}^{\mu\nu} \gamma_{\mu\nu}^a = 2\tilde{R}^{-1} \tilde{g}^{ab} \tilde{\nabla}_b \tilde{R} \quad (4.29)$$

代入场方程得到:

$$\square \tilde{R} = 0 \quad (4.30)$$

这说明 $\Omega = R$ 的受限外尔变换保持标量曲率的调和; 以及

$$\begin{aligned} & 2\tilde{R}^2 \tilde{R}_{\mu\nu} - 2\tilde{R} \tilde{\nabla}_\mu \tilde{\nabla}_\nu \tilde{R} + 2 \left(\delta_\mu^a \tilde{\nabla}_\nu \tilde{R} + \delta_\nu^a \tilde{\nabla}_\mu \tilde{R} - \tilde{g}_{\mu\nu} \tilde{g}^{ab} \tilde{\nabla}_b \tilde{R} \right) \tilde{\nabla}_a \tilde{R} \\ & + 2\tilde{g}_{\mu\nu} \tilde{\nabla} \tilde{R} \cdot \tilde{\nabla} \tilde{R} - \left(\frac{1}{2} \tilde{R} - \Lambda \tilde{R}^3 \right) \tilde{R}^2 \tilde{g}_{\mu\nu} - 4\tilde{\nabla}_\mu \tilde{R} \tilde{\nabla}_\nu \tilde{R} = \kappa \tilde{R}^3 T_{\mu\nu} \end{aligned} \quad (4.31)$$

$\Rightarrow$

$$2\tilde{R}^2 \tilde{R}_{\mu\nu} - 2\tilde{R} \tilde{\nabla}_\mu \tilde{\nabla}_\nu \tilde{R} - \left(\frac{1}{2} \tilde{R} - \Lambda \tilde{R}^3 \right) \tilde{R}^2 \tilde{g}_{\mu\nu} = \kappa \tilde{R}^3 T_{\mu\nu}$$

此式中的能动张量 T 仍然是作用量对原度规做变分的结果, 然而如果不给出物质场的具体形式, 根本无法对其进行处理. 考虑电磁场的能动张量为

$$\begin{aligned} T_{\mu\nu} &= \frac{1}{\mu_0} \left(F_{\mu\rho} F_{\nu}^{\rho} - \frac{1}{4} g_{\mu\nu} F_{\rho\sigma} F^{\rho\sigma} \right) \\ &= \frac{1}{\mu_0} \left(g^{a\rho} F_{\mu\rho} F_{\nu a} - \frac{1}{4} g_{\mu\nu} g^{ab} g^{cd} F_{ac} F_{bd} \right) \\ &= \frac{1}{\mu_0} \tilde{R}^{-2} \left(\tilde{g}^{a\rho} F_{\mu\rho} F_{\nu a} - \frac{1}{4} \tilde{g}_{\mu\nu} \tilde{g}^{ab} \tilde{g}^{cd} F_{ac} F_{bd} \right) \\ &\equiv \tilde{R}^{-2} \tilde{T}_{\mu\nu} \end{aligned} \quad (4.32)$$

这里能动张量在受限外尔变换下只多了一个因子, 原因是电磁场张量 $F_{\mu\nu}$ 是不依赖于联络的. 别的场未必是这样. 将此式代入方程, 得到

$$2\tilde{R} \tilde{R}_{\mu\nu} - 2\tilde{\nabla}_\mu \tilde{\nabla}_\nu \tilde{R} - \left(\frac{1}{2} \tilde{R} - \Lambda \tilde{R}^3 \right) \tilde{R} \tilde{g}_{\mu\nu} = \kappa \tilde{T}_{\mu\nu} \quad (4.33)$$

忽略 $\mathcal{O}(\tilde{R}^4)$ 则有

$$2\tilde{R} \tilde{R}_{\mu\nu} - 2\tilde{\nabla}_\mu \tilde{\nabla}_\nu \tilde{R} - \frac{1}{2} \tilde{R}^2 \tilde{g}_{\mu\nu} = \kappa \tilde{T}_{\mu\nu} \quad (4.34)$$

这个形式上与(4.5)相同的方程可以看成 $f(\tilde{R}) = \tilde{R}^2$ 理论的场方程, 只不过这时 $\tilde{R}$ 很小.

评论: 这里做的事情是给出大 R 极限下的 R^2 理论和小 R 极限下的 R^2 理论存在对应关系. 根据这个关系, 我们立马能看出 $R = 0$ 的解在 R^2 理论中的特殊性, 不严谨地说, 它对应着 R 无穷大的理论, 那当然是一个紫外能标的引力理论, 在本文的后面我们给出了 $R \neq 0$ 的理论实际上等价于 GR, 因此 R^2 理论的新物理全在 $R = 0$ 的解上.

4.1.1 弱场极限

下面使用第二节的弱场方法来解这个方程. 弱场极限下

$$\tilde{g}_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \quad (4.35)$$

$$\tilde{\Gamma}_{\mu\nu}^{\rho} \sim \frac{1}{2}\eta^{\rho\lambda}(\partial_{\mu}h_{\nu\lambda} + \partial_{\nu}h_{\mu\lambda} - \partial_{\lambda}h_{\mu\nu}) \quad (4.36)$$

$$\tilde{R}_{\mu\nu} \sim \frac{1}{2}(\partial_{\mu}\partial^{\rho}h_{\rho\nu} + \partial_{\nu}\partial^{\rho}h_{\rho\mu} - \partial_{\mu}\partial_{\nu}h - \partial_{\rho}\partial^{\rho}h_{\mu\nu}) \quad (4.37)$$

$$\tilde{R} \sim \eta^{\mu\nu}\tilde{R}_{\mu\nu} = \partial^{\mu}\partial^{\nu}h_{\mu\nu} - \partial^{\mu}\partial_{\mu}h \quad (4.38)$$

左式省略 $\mathcal{O}(h^3)$, 则

$$\partial_{\mu}\partial_{\nu}\tilde{R} = -\frac{1}{2}\kappa\tilde{T}_{\mu\nu} \quad (4.39)$$

$$\Rightarrow \partial_{\mu}\partial_{\nu}(\partial^a\partial^bh_{ab} - \partial^a\partial_a h) = -\frac{\kappa}{2\mu_0}\left(F_{\mu a}F_{\nu}^a - \frac{1}{4}\eta_{\mu\nu}F_{ab}F^{ab}\right) \quad (4.40)$$

取规范

$$\partial^a h_{ab} = 0 \quad (4.41)$$

则

$$\partial_{\mu}\partial_{\nu}\square h = \frac{\kappa}{2\mu_0}\left(F_{\mu a}F_{\nu}^a - \frac{1}{4}\eta_{\mu\nu}F_{ab}F^{ab}\right) \quad (4.42)$$

同时要满足麦克斯韦方程

$$\partial^{\mu}F_{\mu\nu} = 0 \quad (4.43)$$

4.1.2 R^2 引力与爱因斯坦引力等价

在作用量

$$S_G = \frac{1}{2\kappa} \int d^4x \sqrt{g} R^2 \quad (4.44)$$

中按如下方式加入辅助场 ϕ

$$\begin{aligned} S_G &= \frac{1}{2\kappa} \int d^4x \sqrt{g} \left(-(c\phi + R)^2 + R^2 \right) \\ &= \frac{1}{2\kappa} \int d^4x \sqrt{g} \left(-c^2\phi^2 - 2c\phi R \right) \end{aligned} \quad (4.45)$$

得到的作用量对 ϕ 变分取极值将恢复原作用量. 做外尔变换

$$g'_{\mu\nu} = \Omega^2 g_{\mu\nu} \quad (4.46)$$

然后有

$$R(g') = \Omega^{-2} (R(g) - 6\Omega^{-1}\square\Omega) \quad (4.47)$$

为了取消 R 与 ϕ 的耦合, 令 $\Omega = \phi^{\frac{1}{2}}$. 则

$$\begin{aligned} S_G &= \frac{1}{2\kappa} \int d^4x \sqrt{g'} \Omega^{-4} \left(-c^2\phi^2 - 2c\phi (\Omega^2 R(g') + 6\Omega^{-1}\square\Omega) \right) \\ &= \frac{1}{2\kappa} \int d^4x \sqrt{g'} \left(-c^2 - 2cR(g') - 12c\phi^{-3/2}\square\phi^{1/2} \right) \\ &= \frac{-c}{\kappa} \int d^4x \sqrt{g'} \left(\frac{c}{2} + R(g') + 6\phi^{-3/2}\square\phi^{1/2} \right) \end{aligned} \quad (4.48)$$

注意这里的 $\square$ 中的联络仍是原度规的.